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Steinhauer et al.

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(54) **VENTILATION MANAGEMENT SYSTEM**

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Assistant Examiner — Douglas Y Sul

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(Continued)

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G16H 20/40 (2018.01)
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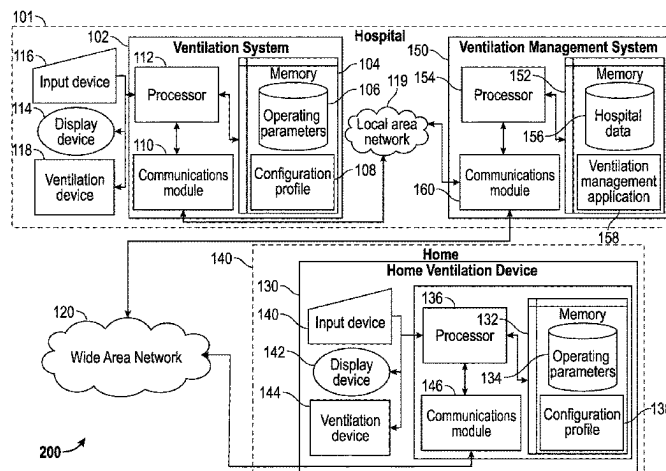
(52) **U.S. Cl.**
CPC **G16H 20/40** (2018.01); **A61M 16/0051**
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(57) **ABSTRACT**

A ventilation management system communicatively couples a remote device such as a mobile device to one or more ventilators for monitoring of ventilator data, including ventilator configuration data, patient physiological statistics, and notifications. When a command to modify a configuration parameter of the ventilator is received while the ventilator is unavailable for communication, the command is cached and provided when the ventilator becomes available. When ventilator becomes available, the system receives an indication that the configuration parameter was modified at the ventilator, an operating condition of the ventilator, and a physiological statistic of a patient associated with the ventilator, the operating condition and physiological statistic having been cached by the ventilator while the ventilator is unavailable. The operating condition of the ventilator, the physiological statistic of the patient, and the indication that the configuration parameter was modified is provided to the remote computing device for display.

24 Claims, 8 Drawing Sheets



Related U.S. Application Data

continuation of application No. 15/394,346, filed on Dec. 29, 2016, now Pat. No. 10,646,674, which is a continuation of application No. 13/830,730, filed on Mar. 14, 2013, now Pat. No. 9,821,129, which is a continuation-in-part of application No. 13/287,419, filed on Nov. 2, 2011, now abandoned.

(51) **Int. Cl.**

A61M 16/10 (2006.01)
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(52) **U.S. Cl.**

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(58) **Field of Classification Search**

CPC **A61M 2205/505**; **A61M 2205/609**; **A61M 2205/3553**; **A61M 2205/6087**; **A61M 2205/6072**; **A61M 2205/6018**; **A61M 2230/435**; **A61M 2230/432**; **A61M 2230/202**

See application file for complete search history.

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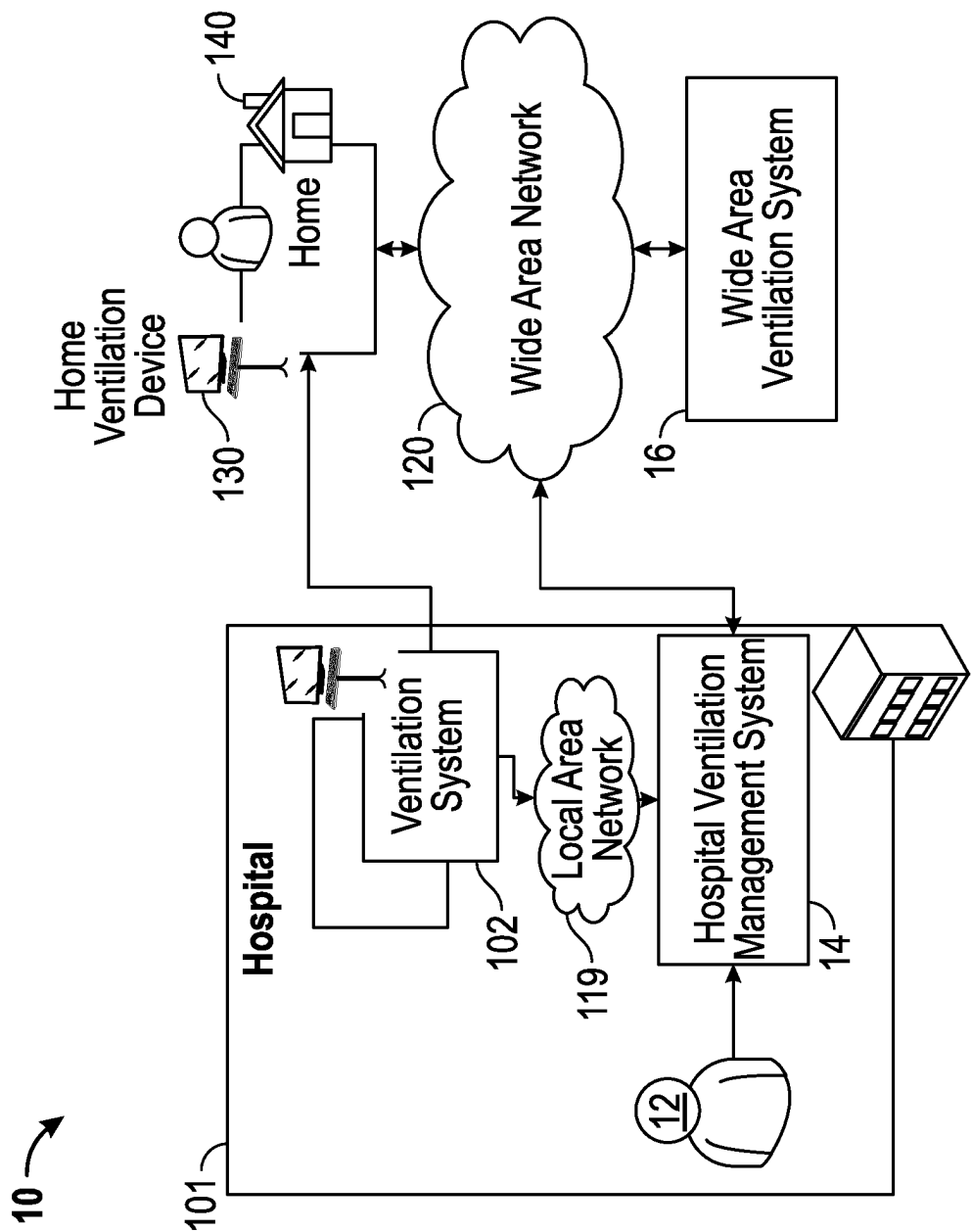


FIG. 1

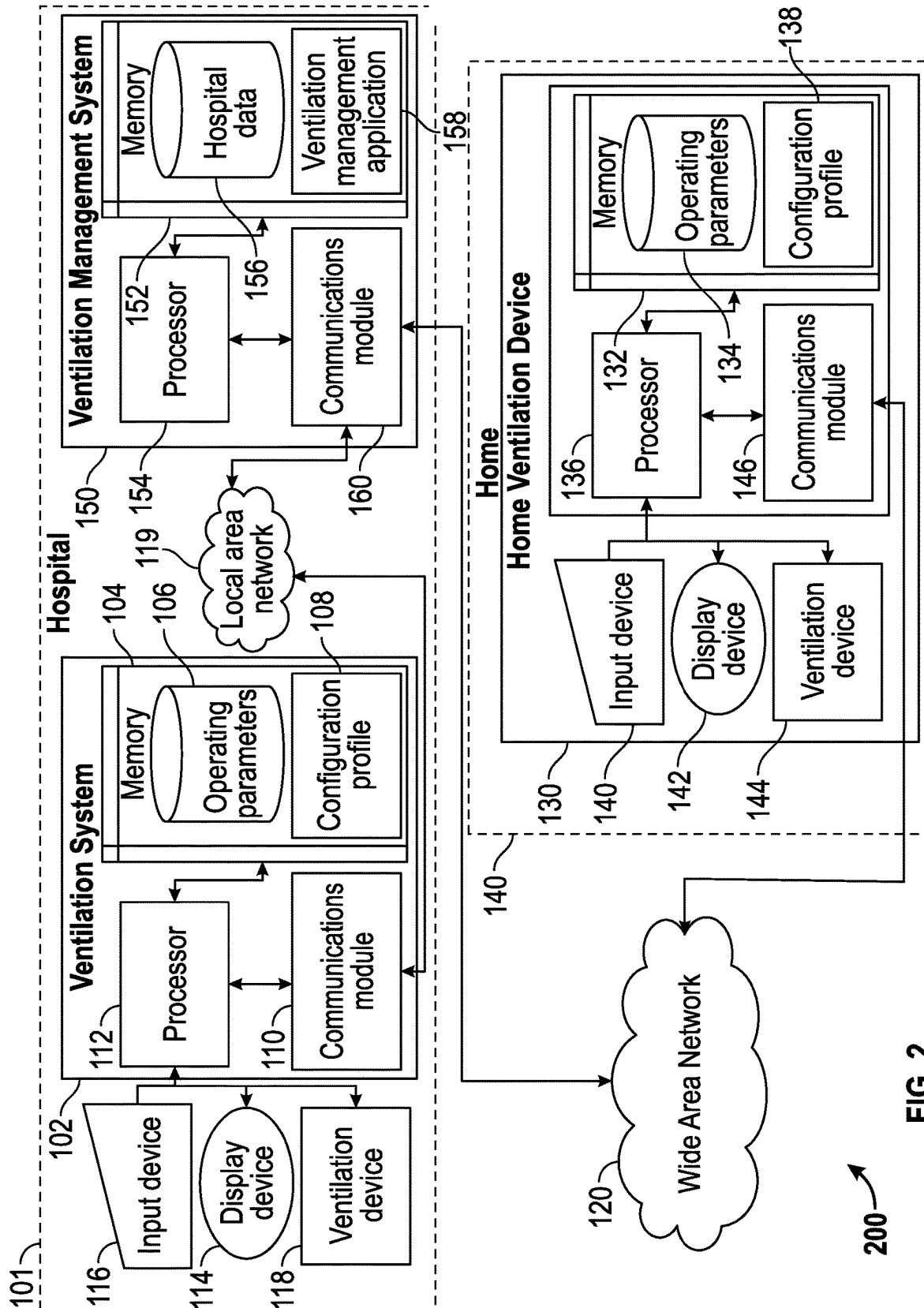


FIG. 2

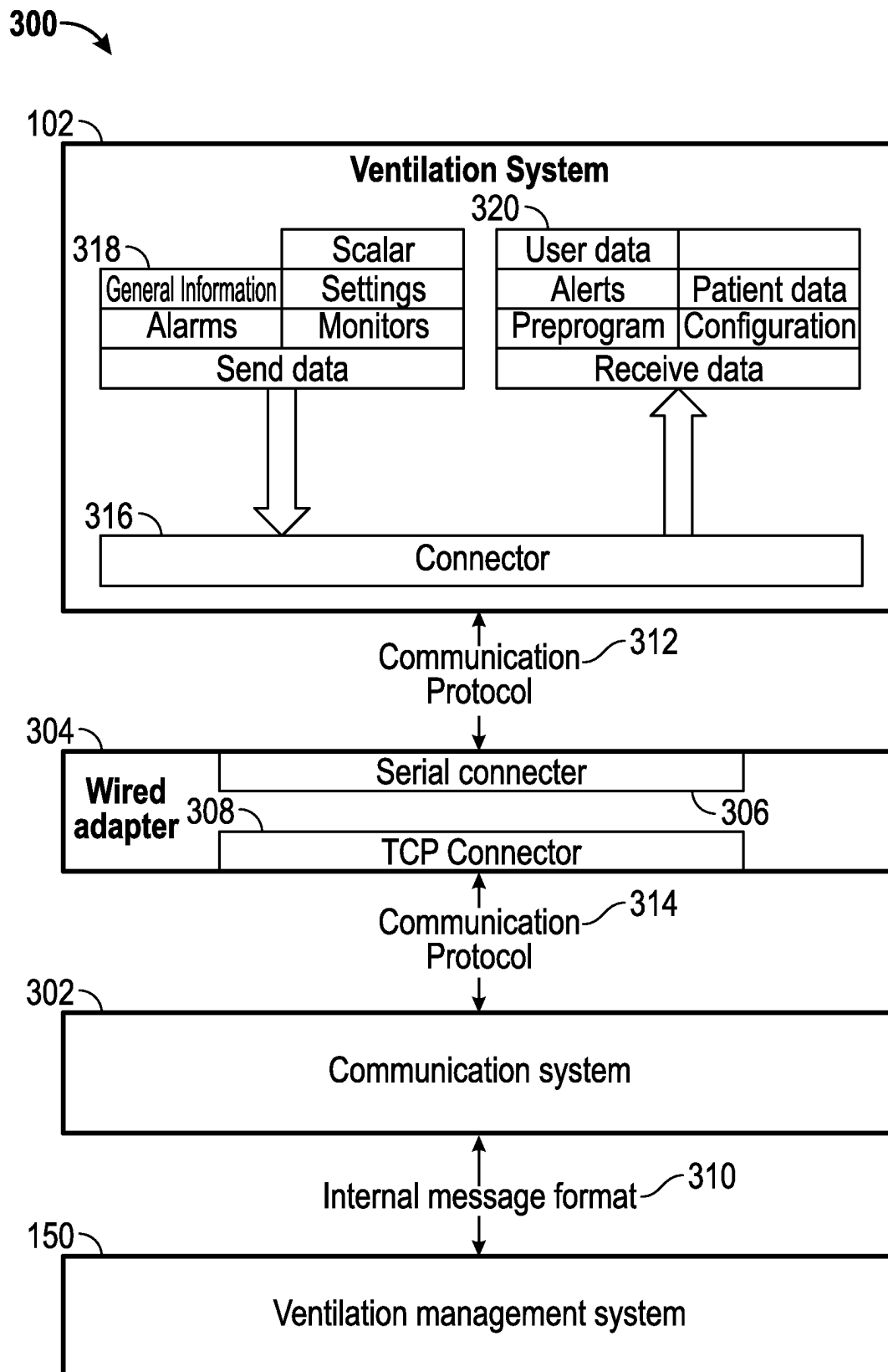


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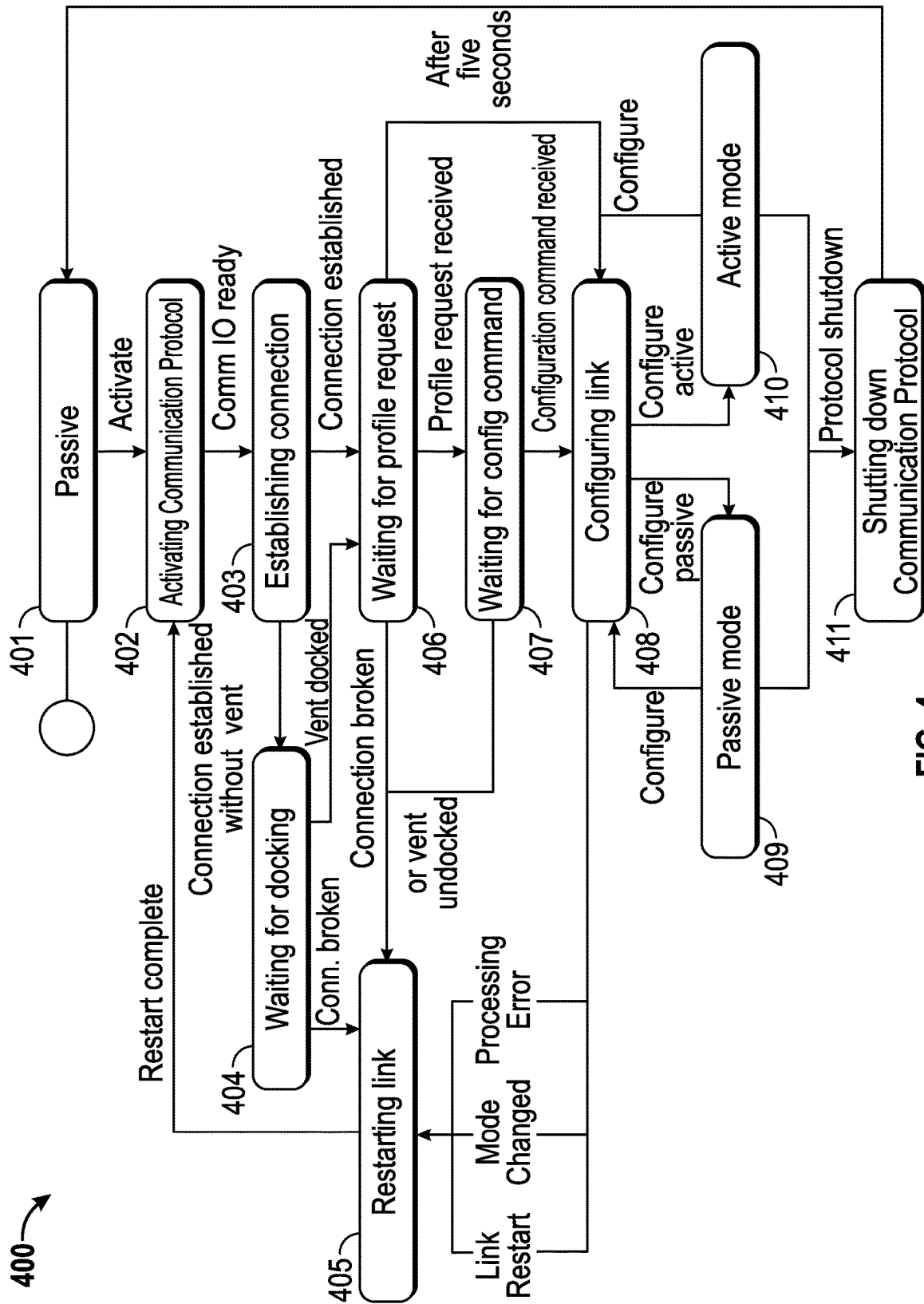


FIG. 4

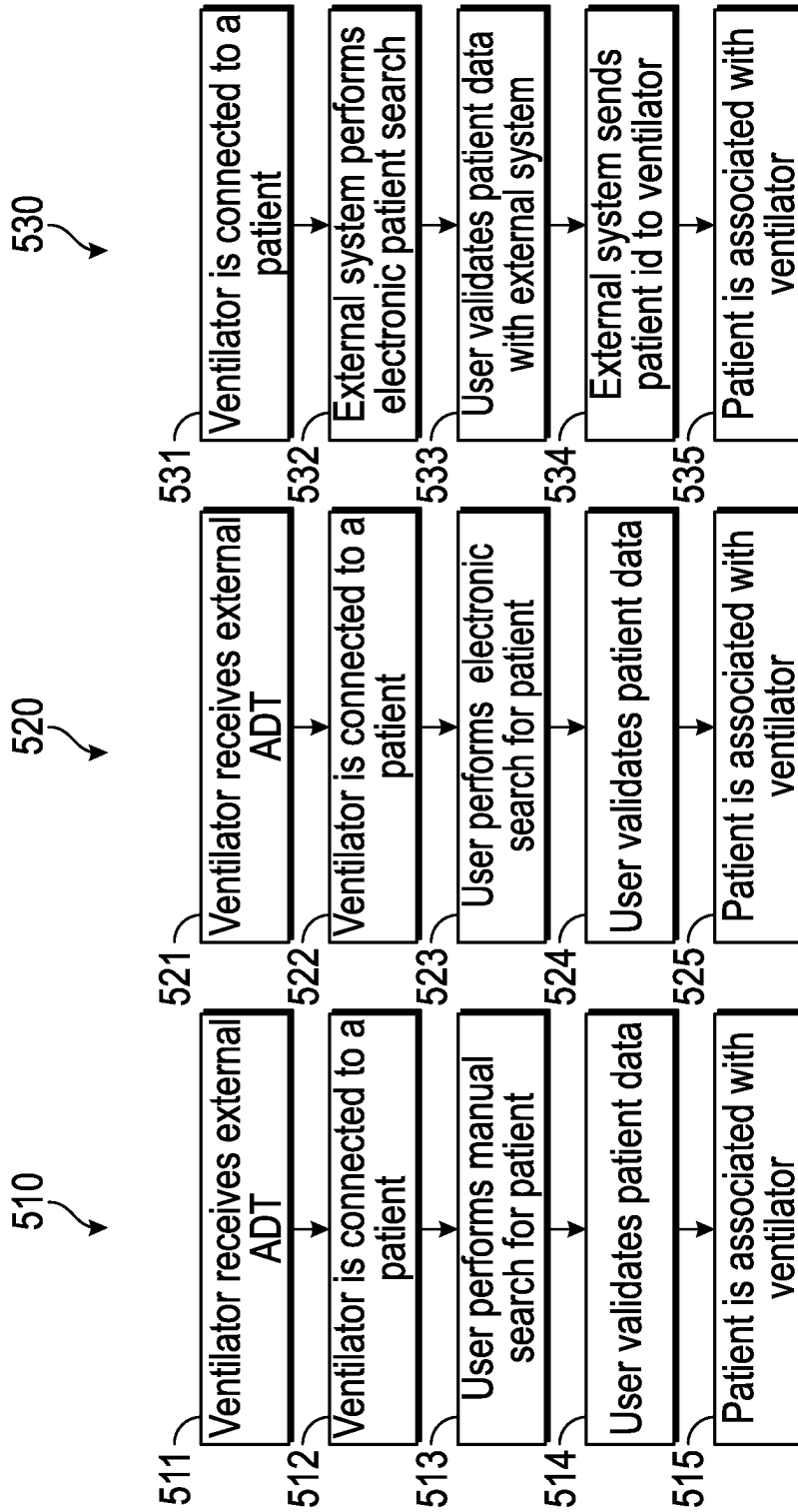


FIG. 5

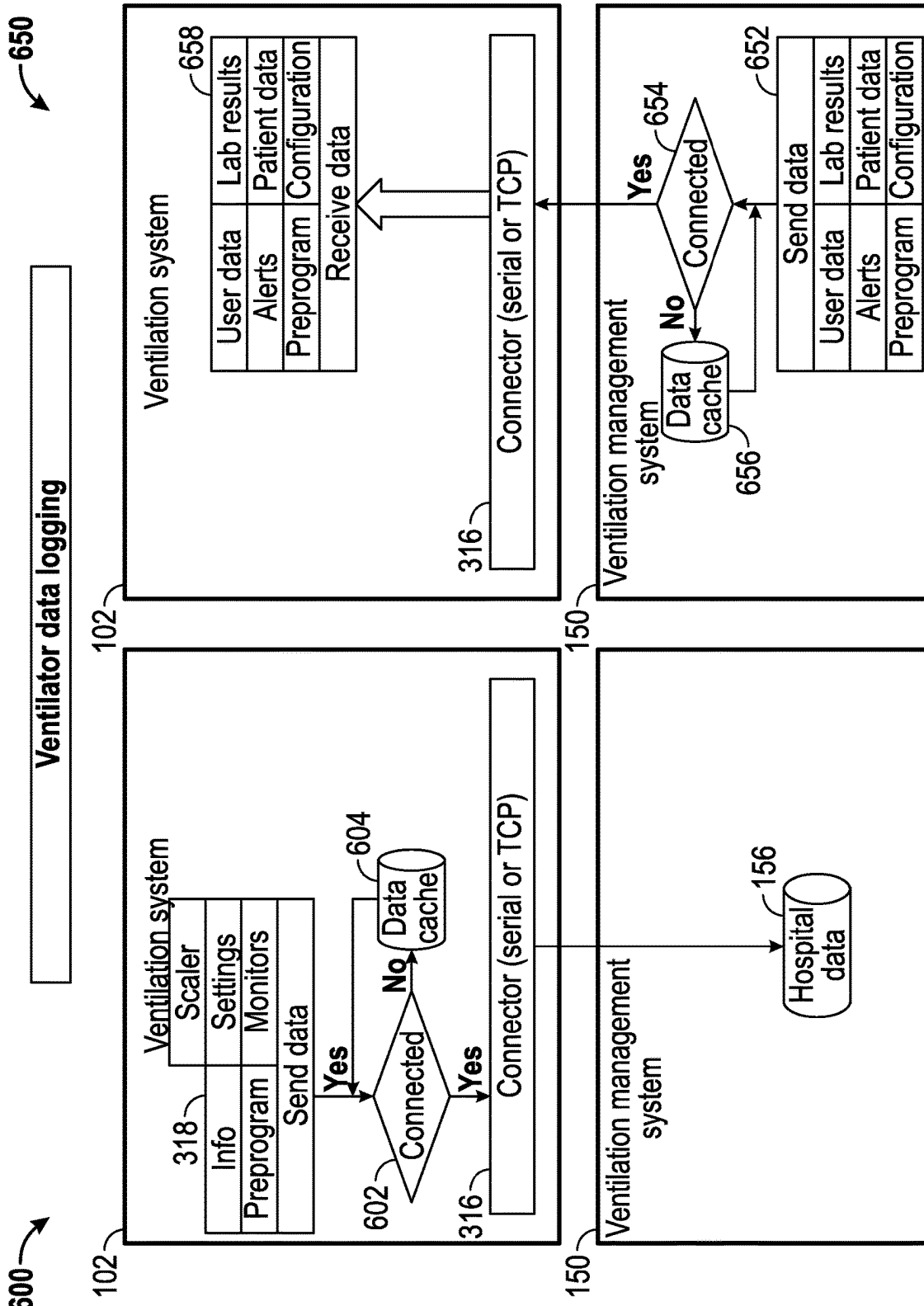


FIG. 6B

FIG. 6A

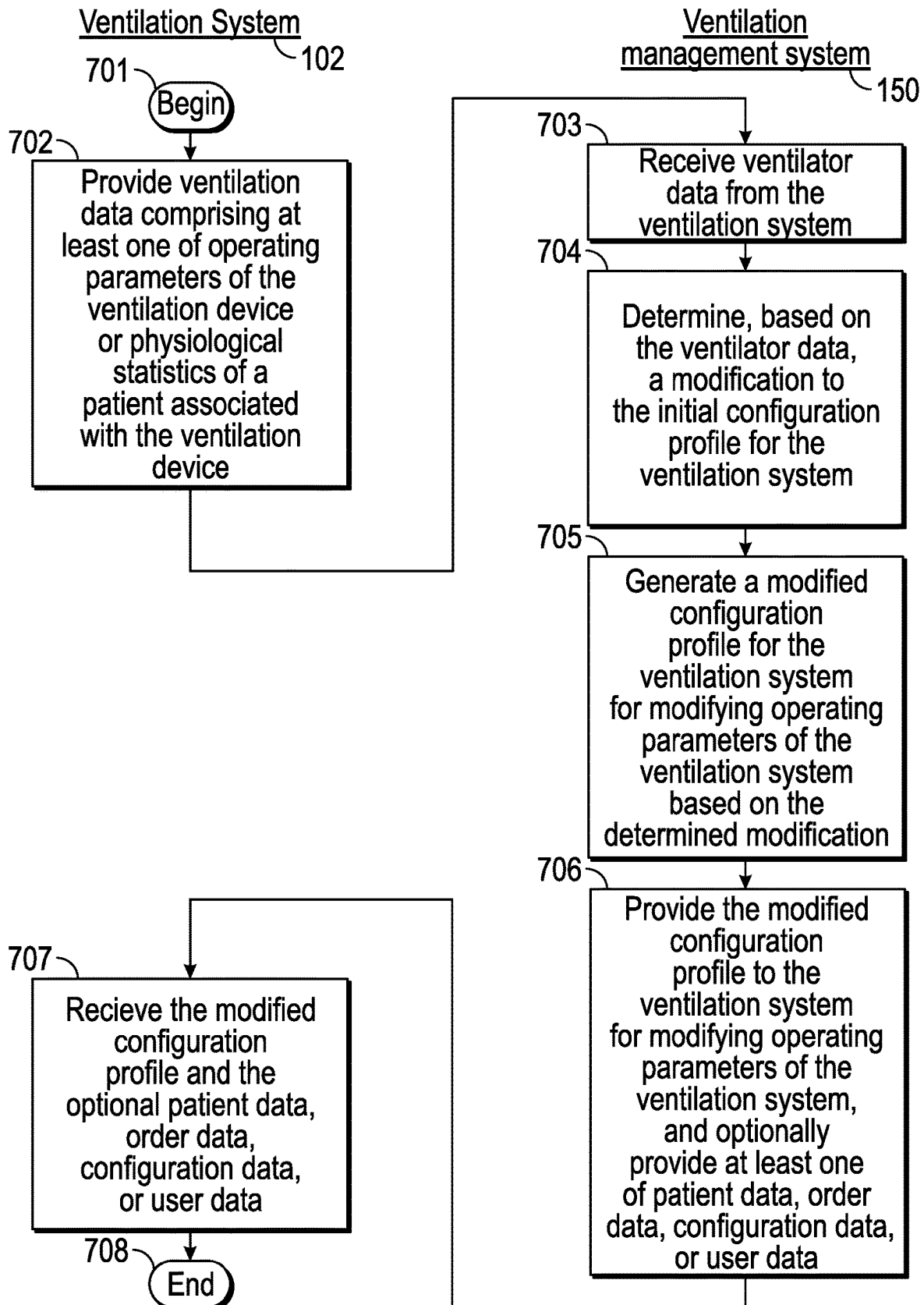


FIG. 7

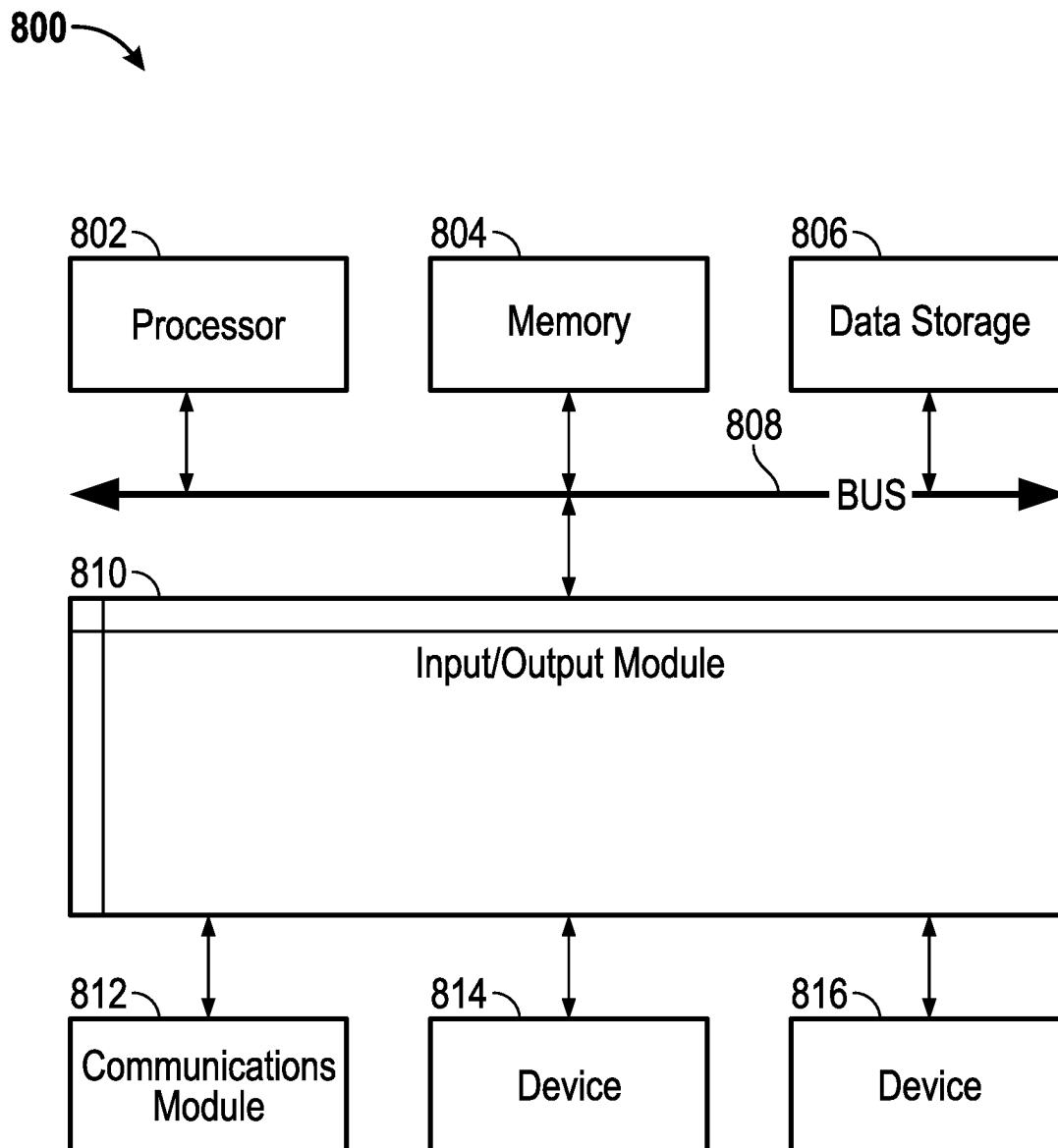


FIG. 8

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VENTILATION MANAGEMENT SYSTEM**CROSS-REFERENCE TO RELATED APPLICATIONS**

The present application is a continuation application of U.S. patent application Ser. No. 16/857,060, entitled "VENTILATION MANAGEMENT SYSTEM", filed on Apr. 23, 2020, now U.S. Pat. No. 11,626,199, which is a continuation application of U.S. patent application Ser. No. 15/394,346, entitled "VENTILATION MANAGEMENT SYSTEM," filed on Dec. 29, 2016, now U.S. Pat. No. 10,646,674, which is a continuation application of U.S. patent application Ser. No. 13/830,730, entitled "VENTILATION MANAGEMENT SYSTEM," filed on Mar. 14, 2013, now U.S. Pat. No. 9,821,129, which is a continuation-in-part application of U.S. patent application Ser. No. 13/287,419, entitled "BI-DIRECTIONAL VENTILATOR COMMUNICATION," filed on Nov. 2, 2011. The disclosures of these applications is hereby incorporated by reference in their entirety for all purposes.

BACKGROUND**Field**

The present disclosure generally relates to medical devices, and more particularly to the configuration of a ventilator.

Description of the Related Art

Medical ventilation systems (or "ventilators," colloquially called "respirators") are machines that are typically used to mechanically provide breathable air or blended gas to lungs in order to assist a patient in breathing. Ventilation systems are chiefly used in intensive care medicine, home care, emergency medicine, and anesthesia. Common ventilation systems are limited to a single direction of communication, and as such are configured to provide information related to the ventilation system for display, but not receive information from a remote source for any purpose to control the ventilator. For example, common ventilation systems send outbound data to another entity, such as a display device, in order to display ventilator settings.

SUMMARY

According to certain embodiments of the present disclosure, a ventilation system is provided. The system includes a ventilation device that is configured to provide breathable air to a patient according to certain operating parameters, a memory that includes instructions, and a processor. The processor is configured to execute the instructions to receive, over a network, at least one of patient data, order data, configuration data, user data, or rules or protocols, and provide a modification of operating parameters of the ventilation device based on the received patient data, order data, configuration data, user data, or rules or protocols.

According to certain embodiments of the present disclosure, a method for configuring a ventilator is provided. The method includes receiving, over a network, at least one of patient data, order data, configuration data, user data, or rules or protocols, and providing a modification of operating parameters of a ventilation device that is configured to provide breathable air to a patient according to the operating

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parameters based on the received patient data, order data, configuration data, user data, or rules or protocols.

According to certain embodiments of the present disclosure, a machine-readable storage medium includes machine-readable instructions for causing a processor to execute a method for configuring a ventilator is provided. The method includes receiving, over a network, at least one of patient data, order data, configuration data, user data, or rules or protocols, and providing a modification of operating parameters of a ventilation device that is configured to provide breathable air to a patient according to the operating parameters based on the received patient data, order data, configuration data, user data, or rules or protocols.

According to certain embodiments of the present disclosure, a ventilator management system is provided. The system includes a memory that includes an initial configuration profile configured to designate operating parameters for a ventilation device, and a processor. The processor is configured to receive ventilator data from the ventilation device, the ventilator data includes at least one of operating parameters of the ventilation device or physiological statistics of a patient associated with the ventilation device, and determine, based on the ventilator data, a modification to the initial configuration profile for the ventilation device. The processor is also configured to generate a modified configuration profile for the ventilation device based on the determined modification.

According to certain embodiments of the present disclosure, a method for managing a plurality of ventilators is provided. The method includes receiving ventilator data from the ventilation device, the ventilator data includes at least one of operating parameters of the ventilation device or physiological statistics of a patient associated with the ventilation device, and determining, based on the ventilator data, a modification to an initial configuration profile for the ventilation device. The method also includes generating a modified configuration profile for the ventilation device based on the determined modification.

According to certain embodiments of the present disclosure, a machine-readable storage medium includes machine-readable instructions for causing a processor to execute a method for managing a plurality of ventilators is provided. The method includes receiving ventilator data from the ventilation device, the ventilator data includes at least one of operating parameters of the ventilation device or physiological statistics of a patient associated with the ventilation device, and determining, based on the ventilator data, a modification to an initial configuration profile for the ventilation device. The method also includes generating a modified configuration profile for the ventilation device based on the determined modification.

It is understood that other configurations of the subject technology will become readily apparent to those skilled in the art from the following detailed description, wherein various configurations of the subject technology are shown and described by way of illustration. As will be realized, the subject technology is capable of other and different configurations and its several details are capable of modification in various other respects, all without departing from the scope of the subject technology. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not as restrictive.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are included to provide further understanding and are incorporated in and

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constitute a part of this specification, illustrate disclosed embodiments and together with the description serve to explain the principles of the disclosed embodiments. In the drawings:

FIG. 1 illustrates an example architecture for a ventilator management system.

FIG. 2 is a block diagram illustrating an example ventilation system, ventilation management system, and home ventilation device from the architecture of FIG. 1 according to certain aspects of the disclosure.

FIG. 3 illustrates an example flow chart of exchanging data between a ventilation system and a ventilation management system.

FIG. 4 illustrates an example flow chart for a communication protocol used by the ventilation system of FIG. 3.

FIG. 5 illustrates example processes for contextualizing ventilator data for a ventilation system.

FIGS. 6A and 6B illustrate example flow charts for caching data on a ventilation system and a ventilation management system.

FIG. 7 illustrates an example process for managing a ventilation system.

FIG. 8 is a block diagram illustrating an example computer system with which the example ventilation system, ventilation management system, and home ventilation device of FIG. 2 can be implemented.

DETAILED DESCRIPTION

In the following detailed description, numerous specific details are set forth to provide a full understanding of the present disclosure. It will be apparent, however, to one ordinarily skilled in the art that the embodiments of the present disclosure may be practiced without some of these specific details. In other instances, well-known structures and techniques have not been shown in detail so as not to obscure the disclosure.

Certain aspects of the disclosed system provide ventilation systems with two-way communication. Specifically, in addition to permitting a ventilation system to output basic ventilation data such as physiological statistics, the disclosed ventilation systems permit output of additional information such as ventilator settings, notifications, patient information, ventilation waveforms, loops or trended data ("scalars"), and ventilation monitoring information. The disclosed ventilation systems also permit input of configuration profiles, rules and clinical protocols, user data, notifications, preprogramming, patient data, and lab results. The disclosed ventilation systems are configured to operate according to the received configuration profiles, rules, and protocols, and in view of the user data, notifications, preprogramming, patient data, and lab results. The data for the ventilation system can also be "contextualized" (e.g., associated with a patient and/or caregiver) using various wired and wireless techniques. The disclosed ventilation systems are configured to provide the output of additional information to, for example, a ventilation management system.

The disclosed ventilation management system is configured to receive the information from one or many ventilation systems, analyze the information, and determine new or modified configuration profiles, rules, and clinical protocols from the received information. The information may be received wired or wirelessly over a network. The disclosed ventilation management system is also configured to provide the new or modified configuration profiles, rules, and clinical protocols back to one or many of the ventilation systems. The ventilation systems managed by the ventilation man-

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agement system can be located either in a healthcare institution (e.g., a hospital) or outside of a healthcare institution (e.g., a home or other care site). Both the ventilation systems and the ventilation management systems are configured to cache data, for example, when the network is not available, so that data may be saved for later transmission.

FIG. 1 illustrates an example architecture 10 for a ventilator management system. The architecture 10 includes a ventilation system 102 and a hospital ventilation management system 14 connected over a local area network (LAN) 119 in a hospital 101, and a home ventilation device 130 in a home 140 connected to a wide area ventilation management system 16 over a wide area network (WAN) 120. The hospital ventilation management system 14, which can be configured, for example, by a clinician 12, other healthcare provider, or administrator, is connected to the wide area ventilation management system 16 through the WAN 120. Furthermore, the home ventilation device 130 may operate substantially similar to, and be configured substantially the same as, the ventilation system 102 of the hospital 101, except that the home ventilation device 130 operates in the home 140.

Each of the ventilation systems 102 is configured to mechanically move breathable air into and out of lungs in order to assist a patient in breathing. The ventilation systems 102 can provide ventilator data, such as notifications, settings, monitor information (e.g., physiological statistics), and scalars to the hospital ventilation management system 14. The ventilation system 102 includes a device having appropriate processor, memory, and communications capabilities for processing and providing ventilator data to the hospital ventilation management system 14. Similarly, the hospital ventilation management system 14 is configured to provide user data, notifications, pre-programmed instructions, lab results, patient data, configuration information, and rules and clinical protocols to each ventilation system 102 in the hospital 101 in order to configure each ventilation system 102 (e.g., remotely over a wired or wireless network, such as LAN 119). The information provided by the hospital ventilation management system 14 to each ventilation system 102 can be based on the information provided to the hospital ventilation management system 14 by each ventilation system 102.

For example, a ventilation system 102 can provide the hospital ventilation management system 14 with a current configuration profile and current monitor information for a patient associated with the ventilation system 102. The hospital ventilation management system 14 can analyze the information provided by the ventilation system 102 in order to determine which modifications, if any, to make to the configuration profile in view of the patient's monitor information. The hospital ventilation management system 14 may then provide a modified configuration profile to the ventilation system 102 so that the ventilation system 102 may treat the patient in accordance with the modified configuration profile.

The hospital ventilation management system 14 is connected to a wide area ventilation management system 16 configured to manage one or many home ventilation devices 130. Although the hospital ventilation management system 14 and the wide area ventilation management system 16 are illustrated as being separate systems, both the hospital ventilation management system 14 and the wide area ventilation management system 16 can be hosted or otherwise executed from a single server. In certain aspects, many servers may share the hosting responsibilities of the hospital ventilation management system 14 and the wide area ven-

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tilation management system 16. The server can be any device having an appropriate processor, memory, and communications capability for hosting the hospital ventilation management system 14 and the wide area ventilation management system 16, and can be in a hospital data center or remotely hosted over a network.

The WAN 120 can include, for example, any one or more of a metropolitan area network (MAN), a wide area network (WAN), a broadband network (BBN), the Internet, and the like. The LAN 119 can include, for example, a personal area network (PAN) or campus area network (CAN). Further, each of the WAN 120 and LAN 119 can include, but is not limited to, any one or more of the following network topologies, including a bus network, a star network, a ring network, a mesh network, a star-bus network, tree or hierarchical network, and the like.

An example use of the ventilator management system will now be provided. A patient associated with the ventilation system 102 is discharged by a clinician 12 from the hospital 101 but still requires ventilation using home ventilation device 130 in the patient's home 140. The hospital ventilation management system 14 registers with the wide area ventilation management system 16, and then sends the patient's information and ventilator information from the ventilation system 102 for the patient to the wide area ventilation management system 16. The home ventilation device 130 is configured using the patient's information and ventilator information and the patient begins treatment using the home ventilation device 130. The clinician monitors the patient's progress with the home ventilation device 130 by reviewing logs from the home ventilation device 130 that are sent to the hospital ventilation management system 14 through the wide area ventilation management system 16. As needed, the clinician may modify the configuration parameters of the home ventilation device 130 remotely by sending new configuration parameters from the hospital ventilation management system 14 to the wide area ventilation management system 16, which then sends the new configuration parameters to the home ventilation device 130 for review by the patient or caregiver. The patient or caregiver accepts the new configuration parameters and the home ventilation device 130 begins to operate according to the new configuration parameters.

FIG. 2 is a block diagram illustrating an example ventilation system 102, ventilation management system 150, and home ventilation device 130 from the architecture 10 of FIG. 1 according to certain aspects of the disclosure. Although the ventilation management system 150 is illustrated as connected to a ventilation system 102 and a home ventilation device 130, the ventilation management system 150 is configured to also connect to infusion pumps, point of care vital signs monitors, and pulmonary diagnostics devices.

The ventilation system 102 is connected to the ventilation management system 150 over the LAN 119 via respective communications modules 110 and 160 of the ventilation system 102 and the ventilation management system 150. The ventilation management system 150 is connected over WAN 120 to the home ventilation device 130 via respective communications modules 160 and 146 of the ventilation management system 150 and the home ventilation device 130. The home ventilation device 130 is configured to operate substantially similar to the ventilation system 102 of the hospital 101, except that the home ventilation device 130 is configured for use in the home 140. The communications modules 110, 160, and 146 are configured to interface with the networks to send and receive information, such as data, requests, responses, and commands to other devices on the

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networks. The communications modules 110, 160, and 146 can be, for example, modems or Ethernet cards.

The ventilation management system 150 includes a processor 154, the communications module 160, and a memory 152 that includes hospital data 156 and a ventilation management application 158. Although one ventilation system 102 is shown in FIG. 2, the ventilation management system 150 is configured to connect with and manage many ventilation systems 102, both ventilation systems 102 for hospitals 101 and home ventilation devices 130 for use in the home 140.

In certain aspects, the ventilation management system 150 is configured to manage many ventilation systems 102 in the hospital 101 according to certain rules and procedures. For example, when powering on, a ventilation system 102 may send a handshake message to the ventilation management system 150 to establish a connection with the ventilation management system 150. Similarly, when powering down, the ventilation system 102 may send a power down message to the ventilation management system 150 so that the ventilation management system 150 ceases communication attempts with the ventilation system 102.

The ventilation management system 150 is configured to support a plurality of simultaneous connections to different ventilation systems 102 and home ventilation devices 130. The number of simultaneous connections can be configured by an administrator in order to accommodate network communication limitations (e.g., limited bandwidth availability). After the ventilation system 102 successfully handshakes with (e.g., connects to) the ventilation management system 150, the ventilation management system 150 may initiate communications to the ventilation system 102 when information becomes available, or at established intervals. The established intervals can be configured by a user so as to ensure that the ventilation system 102 does not exceed an established interval for communicating with the ventilation management system 150.

The ventilation management system 150 can provide the data to the ventilation system 102 in a first-in-first-out (FIFO) order. For instance, if a software upgrade is scheduled to be sent to a ventilation system 102, the software upgrade can be deployed at configurable timeframes in FIFO order for the specified ventilation systems 102. Upon receipt, a ventilation system 102 may initialize the software upgrade on a manual reboot. An admit-discharge-transfer communication can be sent to specified ventilation systems 102 within a certain care area of the hospital 101. A configuration profile communication can be sent to all ventilation systems 102 connected to the ventilation management system 150. On the other hand, orders specific to a patient are sent to the ventilation system 102 associated with the patient.

The ventilation system 102 may initiate a communication to the ventilation management system 150 if an alarm occurs on the ventilation system 102. The alarm may be sent to the beginning of the queue for communicating data to the ventilation management system 150. All other data of the ventilation system 102 may be sent together at once, or a subset of the data can be sent at certain intervals.

The hospital data 156 includes configuration profiles configured to designate operating parameters for the ventilation system 102, operating parameters of the ventilation system 102 and/or physiological statistics of a patient associated with the ventilation system 102. Hospital data 156 also includes patient data for patients at the hospital 101, order (e.g., medication orders, respiratory therapy orders)

data for patients at the hospital **101**, and/or user data (e.g., for caregivers associated with the hospital **101**).

The physiological statistics of the ventilator data includes, for example, a statistic for compliance of the lung (Cdyn, Cstat), flow resistance of the patient airways (Raw), inverse ratio ventilation (I/E), spontaneous ventilation rate, exhaled tidal volume (Vte), total lung ventilation per minute (Ve), peak expiratory flow rate (PEFR), peak inspiratory flow rate (PIFR), mean airway pressure, peak airway pressure, an average end-tidal expired CO₂ and total ventilation rate. The operating parameters include, for example, a ventilation mode, a set mandatory tidal volume, positive end respiratory pressure (PEEP), an apnea interval, a bias flow, a breathing circuit compressible volume, a patient airway type (for example endotracheal tube, tracheostomy tube, face mask) and size, a fraction of inspired oxygen (FiO₂), a breath cycle threshold, and a breath trigger threshold.

The processor **154** of the ventilation management system **150** is configured to execute instructions, such as instructions physically coded into the processor **154**, instructions received from software (e.g., ventilation management application **158**) in memory **152**, or a combination of both. For example, the processor **154** of the ventilation management system **150** executes instructions to receive ventilator data from the ventilation system **102** (e.g., including an initial configuration profile for the ventilation system **102**).

FIG. 3 illustrates an example flow chart **300** of exchanging data between the ventilation system **102** and the ventilation management system **150**. As illustrated in the flow chart **300**, the ventilation system **102** is configured to send ventilator information, notifications (or “alarms”), scalars, operating parameters **106** (or “settings”), physiological statistics (or “monitors”) of a patient associated with the ventilation system **102**, and general information. The notifications include operational conditions of the ventilation system **102** that may require operator review and corrective action. The scalars include parameters that are typically updated periodically (e.g., every 500 ms) and can be represented graphically on a two-dimensional scale. The physiological statistics represent information that the ventilation system **102** is monitoring, and can dynamic based on a specific parameter. The operating parameters **106** represent the operational control values that the caregiver has accepted for the ventilation system **102**. The general information can be information that is unique to the ventilation system **102**, or that may relate to the patient (e.g., a patient identifier). The general information can include an identifier of the version and model of the ventilation system **102**.

In the example of FIG. 3, the data is sent via a serial connector. The data is sent to a wired adapter **304** having a serial connector and a TCP connector **308**. The data is sent using any appropriate communication protocol **400** (e.g., VOXP protocol). FIG. 4 illustrates an example flow chart for a communication protocol, the VOXP protocol, used by the ventilation system **102** of FIG. 3.

The communication protocol **400** of FIG. 4 is configured, in certain aspects, to operate in an active mode and a passive mode. In active mode, the ventilation system **102** both responds to requests (e.g., from the ventilation management system **150**), as well as automatically sends data as it becomes available to the ventilation system **102**. In passive mode, the ventilation system **102** responds to requests but does not automatically send data as it becomes available. The protocol **400** begins by transition from a dormant (or “passive”) mode **401** to starting the VOXP protocol **402** (e.g., to enter into active mode). When the communication input/output port is ready, a connection is established **403**

with the destination (e.g., wired adapter **304**). If the connection is established without a ventilation device **118** being connected to the ventilation system **102**, then the protocol instructs to wait for docking **404** (e.g., of a ventilation device **118**). If a connection is broken while waiting for docking, the link between the ventilation system **102** and the destination is reestablished **405**. Otherwise, when a ventilation device **118** is docked, or a connection is established, the protocol waits for a profile or other data request **406** (e.g., from the ventilation management system **150**). If the connection is broken while waiting for the profile request, the link between the ventilation system **102** and the destination is reestablished. When the profile request is received, ventilation system **102** sends a configuration profile **108** (specifying the capabilities of the ventilation system **102** and the set of operating parameters and other data that it can provide), and then the protocol waits for a configuration command **407** (e.g., from the ventilation management system **150**). When the configuration command is received, a link is established with the destination and the link is configured **408**. If while configuring the link there is a processing error, a mode changes, or the link is restarted, the link is again reestablished **405**. Otherwise, upon configuring the link **408**, the protocol for the ventilation system **102** may enter a passive mode **409** or active mode **410**. In passive mode **409**, the ventilation management system **150** sends requests, at intervals determined by the ventilation management application **158**, for specified information. At each such request, the ventilation system **102** responds with the specified information **318**, which may include notifications (or “alarms”), scalars, operating parameters **106** (or “settings”), and physiological statistics (or “monitors”) of a patient associated with the ventilation system **102**. In active mode **410**, the ventilation system **102** sends specified information **318**, which may include notifications (or “alarms”), scalars, operating parameters **106** (or “settings”), and physiological statistics (or “monitors”) of a patient associated with the ventilation system **102**, as each item becomes available. For example an operating parameter **106** is sent when a user of the ventilation system makes a change to a set value. When the ventilation system is turned off, the protocol signals a shutdown **411**. Upon shutting down, the protocol can automatically enter a dormant mode **401** (e.g., after 5 seconds).

Returning to FIG. 3, the wired adapter **304** is configured to receive **312** the data according to the communication protocol **400** of FIG. 4, and convert the data from a serial connection format to a TCP connection format. The wired adapter **304** then provides **314** the data in the TCP connection format according to the communication protocol **400** of FIG. 4 to a communication system **302**.

The data is received from the ventilation system **102** through the wired adapter **304** by the communication system **302**. The data may be in a native message format of the ventilation system **102**. The communication system **302** is configured to convert the data into an internal messaging format configured for use with a ventilation management system **150**. The conversion can take place according to the system and method of converting messages being sent between data systems using different communication protocols and message structures described in U.S. patent application Ser. No. 13/421,776, entitled “Scalable Communication System,” and filed on Mar. 15, 2012, the disclosure of which is hereby incorporated by reference in its entirety for all purposes. The communication system **302** can include, for example, an interface module for communicating with the wired adapter **304**.

The interface module can include information on the communication protocol **400** (e.g., VOXP protocol) and data structure used by the ventilation system **102** and is configured to both receive messages from and transmit messages between the ventilation system **102** and the ventilation management system **150**. For example, the ventilation management system **150** is configured to provide, through the communication system **302** and the wired adapter **304**, patient data, order data, configuration data, user data, pre-programmed information, vital sign information, rules, notifications, and clinical protocols to the ventilation system **102**. The patient data includes, for example, admit-discharge-transfer data, allergy data, diagnosis data, medication history, procedure history, a patient's name, the patient's medical record number (MRN), lab results, or the patient's visit number. Medication history may include a list of the medications and doses that have been administered to the patient, for example sedative medications, muscle paralytic medications, neural block medications, anti-inflammatory medications. Procedure history may include a list of surgical or other interventional procedures that have been administered, for example cardiothoracic surgery; lung lavage; maxillofacial surgery; chest physiotherapy. The order data includes, for example medication order information, procedure order information for at least one of physical therapy or percussion therapy, sedation order information indicating sedation vacations or modes of ventilator therapy, therapy order information for invasive or non-invasive ventilator therapy, or trial order information for spontaneous breathing trials. The configuration data includes, for example, a patient profile, a user interface configuration, a limit configuration, a notification configuration, or a clinical protocol configuration. The notification configuration can indicate whether certain limits or alerts should be enabled or disabled, and the clinical protocol configuration can be used in a particular area of the hospital **101** (e.g., ICU) and indicate which clinical protocol library should be enabled. A clinical protocol library may include several clinical protocols that may be applicable to a specified group of patients, for example a spontaneous breathing trial clinical protocol. A clinical protocol may include a set of rules defining actions that the ventilation system **102** should effect in response to events such as a change in patient physiological data, for example a spontaneous breathing trial clinical protocol may include a rule that recommences mandatory ventilation in the event that the patient's rapid shallow breathing index (RSBI) exceeds a set threshold. As another example the spontaneous breathing trial clinical protocol may include a rule that a notification should be provided on display device **114** when the patient has been controlling their own respiration within specified limits for a period of one hour. In certain aspects, the notifications can be generated by the ventilation management system **150** and sent to the ventilation system **102** to alert a caregiver or patient near the ventilation system **102**. The user data includes, for example, an identification of a caregiver or a healthcare institution.

After receiving the ventilator data from the ventilation system **102**, the processor **154** of the ventilation management system **150** is configured to determine, based on the ventilator data, a modification to the initial configuration profile for the ventilation system **102**. In certain aspects, the initial configuration profile is received by the ventilation management system **150** from the ventilation system **102**. The processor **154** of the ventilation management system **150** is further configured to generate a modified configuration profile for the ventilation system **102** based on the determined modification. In certain aspects, the modification

to the configuration profile is also determined based on the initial configuration profile of the ventilation system **102**. For example, if the initial configuration profile indicated an average end tidal CO₂ level that was considered clinically too low for the patient, the configuration profile could be modified to increase the average end tidal CO₂.

In certain aspects, the modification to the configuration profile is also determined based on comparing the physiological statistics of the patient with historical patient data (e.g., stored in the hospital data **156**) to identify a modification to at least one operating parameter of the initial configuration profile, and modify the operating parameter based on the identification. For example, if an apnea interval that, based on historical patient data for many patients at the hospital **101**, was not likely to improve the condition of the patient, then the apnea interval of the configuration profile could be modified by the ventilation management system **150**. As another example, if a specified level of tidal ventilation normalized to patient weight, based on historical patient data for many patients at the hospital **101** with a specified diagnosis, has been associated with a reduced length of hospital stay, then the configuration profile could be modified to adjust pressure support to target this level of tidal ventilation.

The processor **154** of the ventilation management system **150** can be further configured to provide the modified configuration profile to the ventilation system **102** for modifying operating parameters **106** in the memory **104** of the ventilation system **102**. The modified configuration profile **108** is stored in the memory **104** of the ventilation system, and used by the processor **112** of the ventilation system **102** to modify the operating parameters **106** in the memory **104** of the ventilation system. In certain aspects, details regarding the modified configuration profile (e.g., the modifications made to operating parameters, an identification of a clinician responsible for approving the modifications, etc.) are provided for display using the display device **114** of the ventilation system **102**.

The ventilation system **102** includes a processor **112**, the communications module **110**, and a memory **104** that includes operating parameters **106** and a configuration profile **108**. The ventilation system **102** also includes an input device **116**, such as a keyboard, scanner, or mouse, an output device **214**, such as a display, and a ventilation device **118** configured to mechanically move breathable air into and out of lungs in order to assist a patient in breathing according to instructions from the ventilation system **102**. The configuration profile **108** includes one or many configuration profiles for operating the ventilation device **118** of the ventilation system **102**. For example, the configuration profile **108** can include a profile for operating the ventilation device **118** in an intensive care unit, neonatal intensive care unit, or surgical room, or a profile for operating the ventilation device **118** for patients with a specified respiratory diagnosis, such as ARDS, neuromuscular disease, pneumonia, or post-surgical recovery.

The processor **112** of the ventilation system **102** is configured to execute instructions, such as instructions physically coded into the processor **112**, instructions received from software (e.g., from configuration profile **108**) in memory **104**, or a combination of both. For example, the processor **112** of the ventilation system **102** executes instructions to configure the ventilation device **118**. The processor **112** of the ventilation system **102** executes instructions from the configuration profile **108** causing the processor **112** to receive, over the LAN **119**, at least one of patient data, order data, configuration data, or user data. The con-

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figuration data can include, for example, an indication (e.g., a set limit) for limiting use of the ventilation system 102 within the hospital 101. The processor 112 of the ventilation system 102 is also configured to provide a modification of operating parameters 106 of the ventilation device 118 based on the received patient data, order data, configuration data, or user data.

In certain aspects the patient data received by the ventilation system 102 includes a patient identifier, such as a MRN, that is obtained through various processes 510, 520, and 530 and used to contextualize data generated by the ventilation system 102 as illustrated in FIG. 5. The contextualization of data includes identifying data generated by a ventilation system 102 as being data associated with a specific patient (a “patient context”). The patient context and ventilation system 102 to patient association can be stored in the memory 103 of the ventilation system 102 or in the hospital data 156 in the memory 152 of the ventilation management system 150.

As provided in process 510 of FIG. 5, a ventilation system 102 can be associated with a patient manually when the ventilation system 102 first receives in step 511 an external admit-discharge-transfer alert (e.g., from the ventilation management system 150 or a hospital information system) for a patient. Next, in step 512, the ventilation system 102 is connected to the patient and in step 513 a caregiver, using input device 116 and display device 114, searches for the patient’s name or identifier (from among a list of patient names/identifiers) on the display device 114 of the ventilation system 102. The patient’s identifier can be found, for example, using a search by care area, patient type, alphabetically, or a list of patients associated with the caregiver. In step 514, the user validates the patient data (e.g., selects the patient to associate with the ventilation system 102) and in step 515 the patient is associated with the ventilation system 102. In certain aspects, a second identifier can be required, such as a medical record number, in order to validate the patient data.

As provided in process 520 of FIG. 5, a ventilation system 102 can be associated with a patient automatically when the ventilation system 102 again first receives in step 521 an external admit-discharge-transfer alert (e.g., from the ventilation management system 150 or a hospital information system) for a patient and the ventilation system 102 is connected to the patient in step 522. Next, in step 523, a clinician performs an electronic search for the patient by, for example, scanning a barcode on the patient’s wrist with the input device 116 or having the ventilation system 102 identify the patient using a radio frequency identification (RFID). Next, in step 524, the user validates the patient data (e.g., confirms the automatically identified patient) and in step 525 the patient is associated with the ventilation system 102.

As provided in process 530 of FIG. 5, a ventilation system 102 can also be associated with a patient automatically when the ventilation system 102 is connected to a patient in step 531 and an external system (e.g., a network scanner connected to a server, such as the ventilation management system 150 or an admit-discharge-transfer system) performs a search for the patient (e.g., using RFID). The user in step 533 validates the patient data identified by the external system and the external system sends the patient identification to the ventilation system 102 in step 534. In step 535 the patient is associated with the ventilation system 102. As yet another example, a ventilator may first be connected to a patient, the ventilation system 102 or user then performs an electronic search by, for example, and RFID or scanned

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patient barcode, the external system validates patient data, the external system sends patient data to the ventilation system 102, and the patient is associated with the ventilation system 102.

In certain aspects, both the ventilation management system 150 and ventilation system 102 are configured to cache data, such as the patient data, order data, configuration data, user data, vital sign information (e.g., physiological statistics of a patient), rules, notifications, clinical protocols, and operating parameters. Cached (or “logged”) data can be used to perform analytics that result in improved patient care. By caching the data even when the ventilation system 102 or the ventilation management system 150 are not connected, the data will have a greater chance of being used for analytics and result in improved patient care. The data may be cached, for example, when the LAN 119 connection is unavailable. The data may then be shared between the ventilation management system 150 and ventilation system 102 when the connection becomes available. Similarly, the data may then be shared between the ventilation management system 150 and ventilation system 102 at regularly scheduled intervals (e.g., every 30 minutes). The scheduled intervals are configurable by a caregiver or other user, and can be based on, for example, the data being transmitted, when a change is made to an operating parameter of the ventilation system 102, or when a measured value, alarm threshold, or monitored value reach a predefined level or rate of change. The home ventilation device 130 can also cache data similar to the ventilation system 102. The data may be cached by the home ventilation device 130, for example, when the WAN 120 connection is unavailable.

For example, any data that is generated by the ventilation system 102 for documentation, clinical decision support, biomedical engineering or maintenance support can be cached in the memory 104 of the ventilation system 102 to be sent out to the ventilation management system 150. Similarly, any data that needs to be sent to the ventilation system 102 from the ventilation management system 150 can be cached in memory 152 at the ventilation management system 150 until a scheduled time to send the data, or a next time the ventilation system 150 and ventilation are connected.

FIGS. 6A and 6B illustrate example flow charts for caching data on a ventilation system 102 and a ventilation management system 150. In FIG. 6A, data 318 for the ventilation system 102, including ventilation system information, alarms, scalars, settings, and monitors, when available, is sent to the ventilation management system 150 via a connector 316 for storage as hospital data 156 when a connection 602 between the ventilation system 102 and the ventilation management system 150 is available. Otherwise, when the connection 602 between the ventilation system 102 and the ventilation management system 150 is not available, the data is stored in a data cache 604 on the ventilation system 102.

In FIG. 6B, data 652 for the ventilation management system 150, including user data, alerts, preprogrammed information, lab results, patient data, and configuration information, when available, is sent to the ventilation system 102 via a connector 316 for storage as data 658 in memory 104 when a connection 654 between the ventilation system 102 and the ventilation management system 150 is available. Otherwise, when the connection 654 between the ventilation system 102 and the ventilation management system 150 is not available, the data is stored in a data cache 656 on the ventilation management system 150.

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FIG. 7 illustrates an example process 700 for managing a ventilation system using the example ventilation system 102 and ventilation management system 150 of FIG. 2. While FIG. 7 is described with reference to FIG. 2, it should be noted that the process steps of FIG. 7 may be performed by other systems.

The process 700 begins by proceeding from beginning step 701 when a ventilation system 102 is initialized and establishes a communication with the ventilation management system 150, to step 702 when the ventilation system 102 provides ventilator data including at least one of operating parameters of the ventilation device 118 or physiological statistics of a patient associated with the ventilation device 118 to the ventilation management system 150. In step 703, the ventilation management system 150 receives the ventilator data from the ventilation system 102 and in step 704 determines, based on the ventilator data, a modification to the initial configuration profile 108 for the ventilation system 102. In step 705 a modified configuration profile is generated for the ventilation system 102 based on the determined modification of step 704, and in step 706 the ventilation management system 706 provides the modified configuration profile to the ventilation system 102 for modifying the operating parameters 106 of the ventilation system 102. The ventilation management system 706 may also optionally provide at least one of patient data, order data, configuration data, or user data to the ventilation system 102 in step 706. In step 707, the ventilation system 102 receives the modified configuration profile and optional patient data, order data, configuration data, or user data. The process 700 then ends in step 708.

FIG. 7 sets forth an example process 700 for managing a ventilation system using the example ventilation system 102 and ventilation management system 150 of FIG. 2. An example will now be described using the example process 700 of FIG. 7.

The process 700 begins by proceeding from beginning step 701 when a ventilation system 102 in the hospital 101 is turned on and establishes a communication with the ventilation management system 150, to step 702 when the ventilation system 102 provides operating parameters of the ventilation device 118, physiological statistics of a patient associated with the ventilation device 118, and an initial configuration profile 108 of the ventilation system 102 to the ventilation management system 150. In step 703, the ventilation management system 150 receives the data from the ventilation system 102 and in step 704 determines that the patient's tidal volume has decreased over the last five minutes by 30%, which is an indication of a degradation in the patient's clinical status. The data also indicates the patient's heart rate has increased. The ventilation management system 150 further determines, based on the ventilator data, that the initial configuration profile 108 for the ventilation system 102 should be modified to increase the breath rate parameter. In an alternative example, in step 704 the ventilation management system 150 uses data from other devices such as lab results data 652 including a blood oxygen measurement and a blood carbon dioxide measurement which indicate that the patient is being over-ventilated. The ventilation management system 150 further determines, based on the lab results data, that the initial configuration profile 108 for the ventilation system 102 should be modified to decrease the breath rate parameter. In step 705 the modified configuration profile having the changed breath rate parameter is generated for the ventilation system 102 based on the determined modification of step 704, and in step 706 the ventilation management system 706 provides

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the modified configuration profile to the ventilation system 102 for modifying the operating parameters 106 of the ventilation system 102. The configuration profile 108 of the ventilation system 102 is modified with the modified configuration profile to increase the patient's breath rate, and the process 700 then ends in step 708.

FIG. 8 is a block diagram illustrating an example computer system 800 with which the ventilation system 102, ventilation management system 150, and home ventilation device 130 of FIG. 2 can be implemented. In certain aspects, the computer system 800 may be implemented using hardware or a combination of software and hardware, either in a dedicated server, or integrated into another entity, or distributed across multiple entities.

Computer system 800 (e.g., ventilation system 102, ventilation management system 150, and home ventilation device 130) includes a bus 808 or other communication mechanism for communicating information, and a processor 802 (e.g., processor 112, 154, and 136) coupled with bus 808 for processing information. By way of example, the computer system 800 may be implemented with one or more processors 802. Processor 802 may be a general-purpose microprocessor, a microcontroller, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Programmable Logic Device (PLD), a controller, a state machine, gated logic, discrete hardware components, or any other suitable entity that can perform calculations or other manipulations of information.

Computer system 800 can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them stored in an included memory 804 (e.g., memory 104, 152, and 132), such as a Random Access Memory (RAM), a flash memory, a Read Only Memory (ROM), a Programmable Read-Only Memory (PROM), an Erasable PROM (EPROM), registers, a hard disk, a removable disk, a CD-ROM, a DVD, or any other suitable storage device, coupled to bus 808 for storing information and instructions to be executed by processor 802. The processor 802 and the memory 804 can be supplemented by, or incorporated in, special purpose logic circuitry.

The instructions may be stored in the memory 804 and implemented in one or more computer program products, i.e., one or more modules of computer program instructions encoded on a computer readable medium for execution by, or to control the operation of, the computer system 800, and according to any method well known to those of skill in the art, including, but not limited to, computer languages such as data-oriented languages (e.g., SQL, dBase), system languages (e.g., C, Objective-C, C++, Assembly), architectural languages (e.g., Java, .NET), and application languages (e.g., PHP, Ruby, Perl, Python). Instructions may also be implemented in computer languages such as array languages, aspect-oriented languages, assembly languages, authoring languages, command line interface languages, compiled languages, concurrent languages, curly-bracket languages, dataflow languages, data-structured languages, declarative languages, esoteric languages, extension languages, fourth-generation languages, functional languages, interactive mode languages, interpreted languages, iterative languages, list-based languages, little languages, logic-based languages, machine languages, macro languages, metaprogramming languages, multiparadigm languages, numerical analysis, non-English-based languages, object-

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oriented class-based languages, object-oriented prototype-based languages, off-side rule languages, procedural languages, reflective languages, rule-based languages, scripting languages, stack-based languages, synchronous languages, syntax handling languages, visual languages, with languages, embeddable languages, and xml-based languages. Memory **804** may also be used for storing temporary variable or other intermediate information during execution of instructions to be executed by processor **802**.

A computer program as discussed herein does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, subprograms, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network. The processes and logic flows described in this specification can be performed by one or more programmable processors executing one or more computer programs to perform functions by operating on input data and generating output.

Computer system **800** further includes a data storage device **806** such as a magnetic disk or optical disk, coupled to bus **808** for storing information and instructions. Computer system **800** may be coupled via input/output module **810** to various devices (e.g., ventilation device **118**). The input/output module **810** can be any input/output module. Example input/output modules **810** include data ports such as USB ports. The input/output module **810** is configured to connect to a communications module **812**. Example communications modules **812** (e.g., communications modules **110**, **160**, and **146**) include networking interface cards, such as Ethernet cards and modems. In certain aspects, the input/output module **810** is configured to connect to a plurality of devices, such as an input device **814** (e.g., input device **116**) and/or an output device **816** (e.g., display device **114**). Example input devices **814** include a keyboard and a pointing device, e.g., a mouse or a trackball, by which a user can provide input to the computer system **800**. Other kinds of input devices **814** can be used to provide for interaction with a user as well, such as a tactile input device, visual input device, audio input device, or brain-computer interface device. For example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, tactile, or brain wave input. Example output devices **816** include display devices, such as a LED (light emitting diode), CRT (cathode ray tube), or LCD (liquid crystal display) screen, for displaying information to the user.

According to one aspect of the present disclosure, the ventilation system **102**, ventilation management system **150**, and home ventilation device **130** can be implemented using a computer system **800** in response to processor **802** executing one or more sequences of one or more instructions contained in memory **804**. Such instructions may be read into memory **804** from another machine-readable medium, such as data storage device **806**. Execution of the sequences of instructions contained in main memory **804** causes processor **802** to perform the process steps described herein. One or more processors in a multi-processing arrangement may also be employed to execute the sequences of instructions contained in memory **804**. In alternative aspects, hard-wired circuitry may be used in place of or in combi-

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nation with software instructions to implement various aspects of the present disclosure. Thus, aspects of the present disclosure are not limited to any specific combination of hardware circuitry and software.

Various aspects of the subject matter described in this specification can be implemented in a computing system that includes a back end component, e.g., as a data server, or that includes a middleware component, e.g., an application server, or that includes a front end component, e.g., a client computer having a graphical user interface or a Web browser through which a user can interact with an implementation of the subject matter described in this specification, or any combination of one or more such back end, middleware, or front end components. The components of the system can be interconnected by any form or medium of digital data communication, e.g., a communication network. The communication network (e.g., local area network **119** and wide area network **120**) can include, for example, any one or more of a personal area network (PAN), a local area network (LAN), a campus area network (CAN), a metropolitan area network (MAN), a wide area network (WAN), a broadband network (BBN), the Internet, and the like. Further, the communication network can include, but is not limited to, for example, any one or more of the following network topologies, including a bus network, a star network, a ring network, a mesh network, a star-bus network, tree or hierarchical network, or the like. The communications modules can be, for example, modems or Ethernet cards.

Computing system **800** can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other. Computer system **800** can be, for example, and without limitation, a desktop computer, laptop computer, or tablet computer. Computer system **800** can also be embedded in another device, for example, and without limitation, a mobile telephone, a personal digital assistant (PDA), a mobile audio player, a Global Positioning System (GPS) receiver, a video game console, and/or a television set top box.

The term “machine-readable storage medium” or “computer readable medium” as used herein refers to any medium or media that participates in providing instructions or data to processor **802** for execution. Such a medium may take many forms, including, but not limited to, non-volatile media, volatile media, and transmission media. Non-volatile media include, for example, optical disks, magnetic disks, or flash memory, such as data storage device **806**. Volatile media include dynamic memory, such as memory **804**. Transmission media include coaxial cables, copper wire, and fiber optics, including the wires that comprise bus **808**. Common forms of machine-readable media include, for example, floppy disk, a flexible disk, hard disk, magnetic tape, any other magnetic medium, a CD-ROM, DVD, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, a RAM, a PROM, an EPROM, a FLASH EPROM, any other memory chip or cartridge, or any other medium from which a computer can read. The machine-readable storage medium can be a machine-readable storage device, a machine-readable storage substrate, a memory device, a composition of matter effecting a machine-readable propagated signal, or a combination of one or more of them.

As used herein, the phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each

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member of the list (i.e., each item). The phrase “at least one of” does not require selection of at least one item; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

Furthermore, to the extent that the term “include,” “have,” or the like is used in the description or the claims, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim.

A reference to an element in the singular is not intended to mean “one and only one” unless specifically stated, but rather “one or more.” All structural and functional equivalents to the elements of the various configurations described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and intended to be encompassed by the subject technology. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the above description.

While this specification contains many specifics, these should not be construed as limitations on the scope of what may be claimed, but rather as descriptions of particular implementations of the subject matter. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the aspects described above should not be understood as requiring such separation in all aspects, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

The subject matter of this specification has been described in terms of particular aspects, but other aspects can be implemented and are within the scope of the following claims. For example, the actions recited in the claims can be performed in a different order and still achieve desirable results. As one example, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing may be advantageous. Other variations are within the scope of the following claims.

These and other implementations are within the scope of the following claims.

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What is claimed is:

1. A ventilator management system, comprising:

a memory storing ventilator configuration data, patient physiological statistics, and notifications; and one or more processors configured to:

communicatively connect a ventilator to a remote computing device over a network, the ventilator being configured for communication in a passive mode in which the ventilator responds to requests without automatically sending data as it becomes available; receive a command designated for the ventilator while the ventilator is unavailable for communication with the remote computing device, the command to modify a configuration parameter of the ventilator; cache, in the memory, the command while the ventilator is unavailable and provide the cached command from the memory to the ventilator responsive to the ventilator becoming available for communication with the remote computing device;

receive, from the ventilator when the ventilator becomes available, an indication that the configuration parameter was modified at the ventilator, an operating condition of the ventilator, and a physiological statistic of a patient associated with the ventilator, the operating condition and physiological statistic having been cached by the ventilator while the ventilator is unavailable;

provide, to the ventilator while the ventilator is in the passive mode, requests for first ventilation data associated with the patient at periodic intervals;

receive the first ventilation data from the ventilator responsive to the requests for first ventilation data; and

provide, to the remote computing device for display, the first ventilation data, the operating condition of the ventilator and the physiological statistic of the patient with the indication that the configuration parameter was modified.

2. The ventilator management system of claim 1, wherein the one or more processors are further configured to:

receive an indication that a ventilator is available for communication in the passive mode;

wherein the requests for first ventilation data are provided based on the indication;

switch the ventilator from the passive mode to an active mode wherein the ventilator automatically sends data as it becomes available; and

receive, from the ventilator while the ventilator is in the active mode, second ventilation data as it becomes available at the ventilator without providing the ventilator a request.

3. The ventilator management system of claim 2, wherein in the passive mode, the one or more processors determine data communication intervals for the ventilator by sending the requests for first ventilation data, and

wherein in the active mode, the ventilator determines communication intervals by automatically sending the second ventilation data as it becomes available at the ventilator.

4. The ventilator management system of claim 1, wherein the one or more processors are further configured to:

receive, from the remote computing device, a protocol configuration for a particular care area of a hospital;

select, for the ventilator based on the received protocol configuration, from a plurality of protocol libraries, a

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protocol library applicable to a group of patients associated with the particular care area; and
 cause the ventilator to be configured with a set of rules associated with the selected protocol library, the set of rules defining actions that the ventilator performs in response to a predetermined change in patient physiological data.

5. The ventilator management system of claim 1, wherein the one or more processors are further configured to:
 compare the physiological statistic of the patient with historical patient data;
 identify a modification the configuration parameter of the ventilator based on the comparing; and
 cause the ventilator to modify the configuration parameter according to the modification.

6. The ventilator management system of claim 1, wherein the one or more processors are further configured to:
 receive an indication to set specified limits on a respiration of the ventilator; and
 provide a notification for display when the patient has been controlling the patient's own respiration within specified limits for a predetermined period of time.

7. The ventilator management system of claim 1, wherein the one or more processors are further configured to:
 provide, to the ventilator, before the configuration parameter is modified, an indication that the patient accepts the configuration parameter.

8. The ventilator management system of claim 1, wherein the one or more processors are further configured to, before the physiological statistic of a patient is received:
 receive a patient identifier associated with the patient;
 prompt, after receiving the patient identifier, a clinician for a confirmation to associate the patient with the ventilator;
 receive the confirmation responsive to prompting the clinician; and
 associate the ventilator with the patient based on the confirmation,
 wherein the physiological statistic of a patient is received based on the association of the patient with the ventilator.

9. The ventilator management system of claim 8, wherein the one or more processors are further configured to, before the physiological statistic of a patient is received:
 receive a search for list of patients associated with a care area or a clinician;
 provide the list of patients for display; and
 receive, from the list of patients, a selection of the patient associated with the patient identifier.

10. The ventilator management system of claim 1, wherein the one or more processors are further configured to:
 receive a physiological parameter associated with the patient from the remote computing device;
 determine, based on the physiological parameter, that the patient is not receiving a correct amount of ventilation from the ventilator; and
 cause, responsive to determining that the patient is not receiving the correct amount of ventilation, the ventilator to adjust an operating parameter associated with a breath rate,
 wherein the modified configuration profile is configured to cause the ventilator to adjust an amount of ventilation provided to the patient based on the adjusted operating parameter.

11. The ventilator management system of claim 1, wherein the remote computing device is a mobile device.

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12. A machine-implemented method, comprising:
 communicatively connecting a ventilator to a remote computing device over a network, the ventilator being configured for communication in a passive mode in which the ventilator responds to requests without automatically sending data as it becomes available;
 receiving a command designated for the ventilator while the ventilator is unavailable for communication with the remote computing device, the command to modify a configuration parameter of the ventilator;
 caching, in a memory, the command while the ventilator is unavailable and providing the cached command from the memory to the ventilator responsive to the ventilator becoming available for communication with the remote computing device;
 receiving, from the ventilator when the ventilator becomes available, an indication that the configuration parameter was modified at the ventilator, an operating condition of the ventilator, and a physiological statistic of a patient associated with the ventilator, the operating condition and physiological statistic having been cached by the ventilator while the ventilator is unavailable; and
 provide, to the ventilator while the ventilator is in the passive mode, requests for first ventilation data associated with the patient at periodic intervals;
 receive the first ventilation data from the ventilator responsive to the requests for first ventilation data;
 providing, to the remote computing device for display, the first ventilation data, the operating condition of the ventilator and the physiological statistic of the patient with the indication that the configuration parameter was modified.

13. The machine-implemented method of claim 12, further comprising:
 receiving an indication that a ventilator is available for communication in the passive mode;
 wherein the requests for first ventilation data are provided based on the indication;
 switching the ventilator from the passive mode to an active mode wherein the ventilator automatically sends data as it becomes available; and
 receiving, from the ventilator while the ventilator is in the active mode, second ventilation data as it becomes available at the ventilator without providing the ventilator a request.

14. The machine-implemented method of claim 13, comprising:
 determining data communication intervals for the ventilator by one or more processors external to the ventilator by sending the requests for the first ventilation data, and
 wherein in the active mode, the ventilator determines communication intervals by automatically sending the second ventilation data as it becomes available at the ventilator.

15. The machine-implemented method of claim 12, further comprising:
 receiving, from the remote computing device, a protocol configuration for a particular care area of a hospital;
 selecting, for the ventilator based on the received protocol configuration, from a plurality of protocol libraries, a protocol library applicable to a group of patients associated with the particular care area; and
 causing the ventilator to be configured with a set of rules associated with the selected protocol library, the set of

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rules defining actions that the ventilator performs in response to a predetermined change in patient physiological data.

16. The machine-implemented method of claim 12, further comprising:

comparing the physiological statistic of the patient with historical patient data;
identifying a modification the configuration parameter of the ventilator based on the comparing; and
causing the ventilator to modify the configuration parameter according to the modification.

17. The machine-implemented method of claim 12, further comprising:

receiving an indication to set specified limits on a respiration of the ventilator; and
providing a notification for display when the patient has been controlling the patient's own respiration within specified limits for a predetermined period of time.

18. The machine-implemented method of claim 12, further comprising:

providing, to the ventilator, before the configuration parameter is modified, an indication that the patient accepts the configuration parameter.

19. The machine-implemented method of claim 12, further comprising, before the physiological statistic of a patient is received:

receiving a patient identifier associated with the patient;
prompting, after receiving the patient identifier, a clinician for a confirmation to associate the patient with the ventilator;
receiving the confirmation responsive to prompting the clinician; and
associating the ventilator with the patient based on the confirmation,
wherein the physiological statistic of a patient is received based on the association of the patient with the ventilator.

20. The machine-implemented method of claim 19, further comprising, before the physiological statistic of a patient is received:

receiving a search for list of patients associated with a care area or a clinician;
providing the list of patients for display; and
receiving, from the list of patients, a selection of the patient associated with the patient identifier.

21. The machine-implemented method of claim 12, further comprising:

receiving a physiological parameter associated with the patient from the remote computing device;
determining, based on the physiological parameter, that the patient is not receiving a correct amount of ventilation from the ventilator; and
causing, responsive to determining that the patient is not receiving the correct amount of ventilation, the ventilator to adjust an operating parameter associated with a breath rate,
wherein the modified configuration profile is configured to cause the ventilator to adjust an amount of ventilation provided to the patient based on the adjusted operating parameter.

22. A non-transitory machine-readable storage medium comprising machine-readable instructions that, when

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executed by one or more processors, cause the one or more processors to perform operations comprising:

communicatively connecting a ventilator to a remote computing device over a network, the ventilator being configured for communication in a passive mode in which the ventilator responds to requests without automatically sending data as it becomes available;

receiving a command designated for the ventilator while the ventilator is unavailable for communication with the remote computing device, the command to modify a configuration parameter of the ventilator;

caching, in a memory, the command while the ventilator is unavailable and providing the cached command from the memory to the ventilator responsive to the ventilator becoming available for communication with the remote computing device;

receiving, from the ventilator when the ventilator becomes available, an indication that the configuration parameter was modified at the ventilator, an operating condition of the ventilator, and a physiological statistic of a patient associated with the ventilator, the operating condition and physiological statistic having been cached by the ventilator while the ventilator is unavailable; and

provide, to the ventilator while the ventilator is in the passive mode, requests for first ventilation data associated with the patient at periodic intervals;

receive the first ventilation data from the ventilator responsive to the requests for ventilation data;

providing, to the remote computing device for display, the first ventilation data, the operating condition of the ventilator and the physiological statistic of the patient with the indication that the configuration parameter was modified.

23. The non-transitory machine-readable storage medium of claim 22 comprising machine-readable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising:

receiving an indication that a ventilator is available for communication in the passive mode, wherein the requests for first ventilation data are provided based on the indication;

switching the ventilator from the passive mode to an active mode wherein the ventilator automatically sends data as it becomes available; and

receiving, from the ventilator while the ventilator is in the active mode, second ventilation data as it becomes available at the ventilator without providing the ventilator a request.

24. The non-transitory machine-readable storage medium of claim 23 comprising machine-readable instructions that, when executed by one or more processors, cause the one or more processors to perform operations comprising:

wherein in the passive mode, the one or more processors determine data communication intervals for the ventilator by sending the requests for first ventilation data, and

wherein in the active mode, the ventilator determines communication intervals by automatically sending the second ventilation data as it becomes available at the ventilator.

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